

Supplementary Information

ITAI 1371 – Introduction to Machine Learning | Final Project

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Part B – AI Usage Disclosure

Did you use any AI tools?

Yes. I used Claude (Anthropic) and GitHub Copilot during parts of this project. The table below breaks down exactly where and how.

AI Tool	Sections Used In	Type of Help
Claude (Anthropic)	Report writing, code structure	Draft text and code templates
GitHub Copilot	Jupyter notebook cells	Code autocomplete suggestions
No AI used	Discussion, model design, results analysis	Written and decided independently

Report Writing (Claude)

I used Claude to help draft the opening sections of the written report — specifically the Introduction, Problem Definition, Approach, and Conclusion. I gave it my project proposal and the actual metric outputs, and it produced a starting draft. I then read each paragraph, added my own results and numbers, and rewrote sections that were too generic or didn't match what I actually found in the data.

Code Help (GitHub Copilot)

While writing the Jupyter notebook, Copilot suggested autocomplete for repetitive sklearn code — things like the GridSearchCV setup, metric import statements, and confusion matrix plotting. I looked at every suggestion before using it, ran the cell, checked the output, and only kept code I understood.

What I did without any AI help

The following decisions and writing were entirely my own:

- Choosing to use 33rd and 66th percentile cuts to define popularity tiers
- Selecting which 9 audio features to use for clustering (deliberately leaving out popularity)
- Designing the genre fairness analysis and deciding how to measure prediction bias
- Deciding to compare four models each for regression and classification
- The entire Discussion section (Section 5 of the report)
- Interpreting why linear models had low R^2 and why classification accuracy was near chance

How I handled AI-assisted code

For every cell in the notebook with Copilot-suggested code, I: (1) read it line by line before running it, (2) added my own comments explaining what each part does, (3) ran the cell and verified the output looked right, and (4) could explain the logic if asked. I did not include code I didn't understand.

Part C – Code Logic Diagram

The diagram below shows the full code workflow from data loading to final output. Each box is one major processing step; arrows show the order of execution.

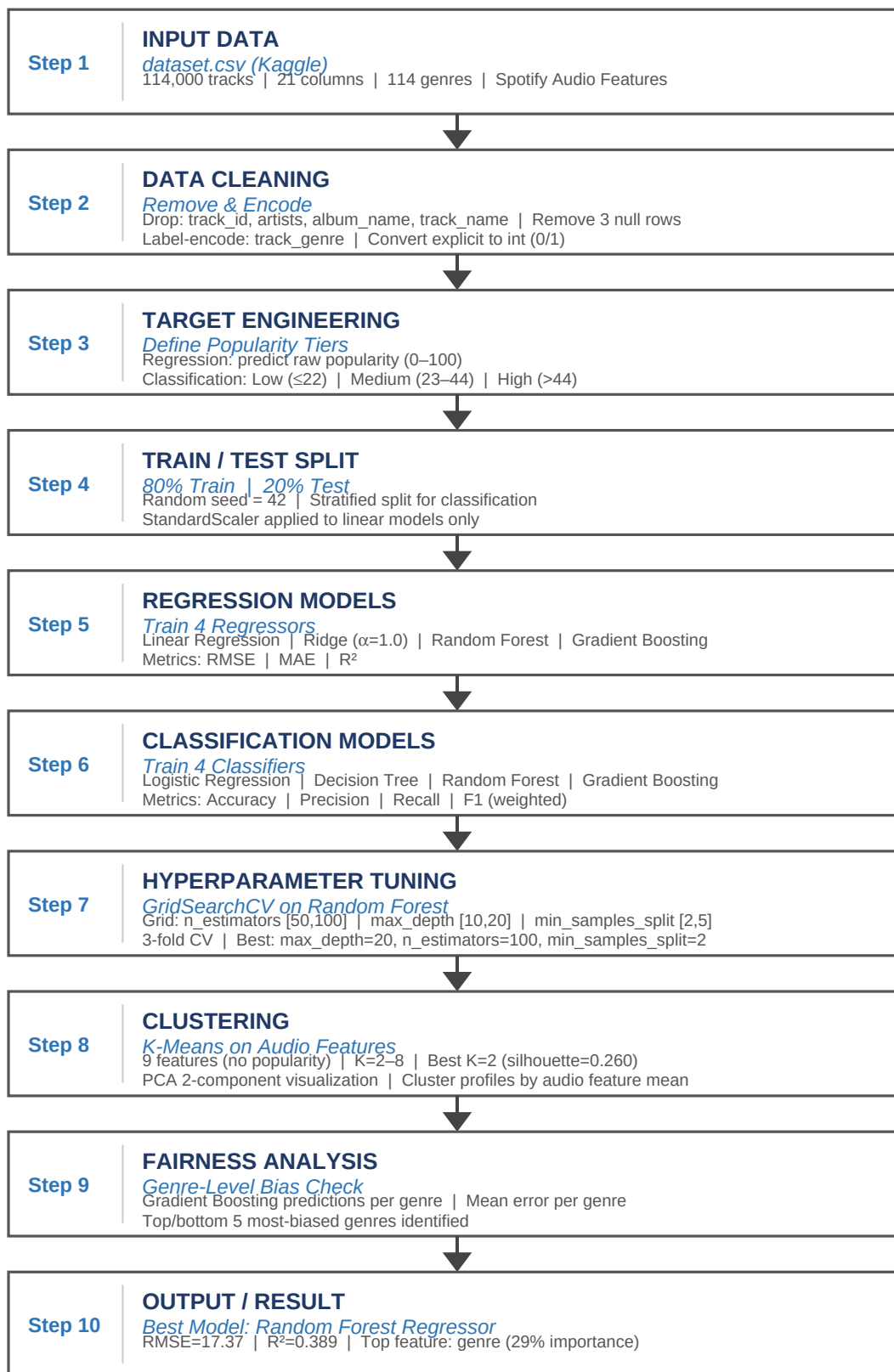


Figure: Full code workflow — from raw CSV input through data cleaning, target engineering, train/test split, regression and classification model training, hyperparameter tuning, K-Means clustering, genre fairness analysis, and final output.